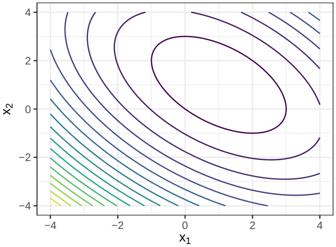
Optimization in Machine Learning

Advanced first order methods
GD on quadratic forms



Learning goals

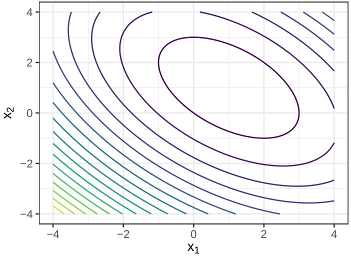
- Eigendecomposition of quadratic forms
- GD steps in eigenspace

Slides based on ▸ Goh 2017

QUADRATIC FORMS & GD

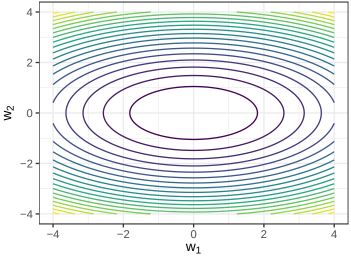
- We consider the quadratic function $q(\mathbf{x}) = \mathbf{x}^T \mathbf{A}\mathbf{x} - \mathbf{b}^T \mathbf{x}$
- We assume that the Hessian $\mathbf{H} = 2\mathbf{A}$ has full rank
- We assume q is convex, so $\mathbf{A}$ is psd
- The optimal solution is $\mathbf{x}^* = \frac{1}{2}\mathbf{A}^{-1}\mathbf{b}$
- With $\nabla q(\mathbf{x}) = 2\mathbf{A}\mathbf{x} - \mathbf{b}$, gradient descent iterates as

$$\mathbf{x}^{[t+1]} = \mathbf{x}^{[t]} - \alpha(2\mathbf{A}\mathbf{x}^{[t]} - \mathbf{b})$$



EIGENDECOMPOSITION OF QUADRATIC FORMS

- We want to work in the coordinate system induced by q
- **Recall:** That system is given by the eigenvectors of $\mathbf{H} = 2\mathbf{A}$
- Eigendecomposition: $\mathbf{A} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^T$
- $\mathbf{V}$ contains eigenvectors $\boldsymbol{v}_i$ and $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$ stores eigenvalues
- Change of basis: $\boldsymbol{w}^{[t]} = \mathbf{V}^T(\boldsymbol{x}^{[t]} - \boldsymbol{x}^*)$



GD STEPS IN EIGENSPACE

- With $\mathbf{w}^{[t]} = \mathbf{V}^T(\mathbf{x}^{[t]} - \mathbf{x}^*)$, a single GD step starts with

$$\mathbf{x}^{[t+1]} = \mathbf{x}^{[t]} - \alpha(2\mathbf{A}\mathbf{x}^{[t]} - \mathbf{b})$$

- This becomes

$$\mathbf{w}^{[t+1]} = \mathbf{w}^{[t]} - 2\alpha\mathbf{\Lambda}\mathbf{w}^{[t]}$$

  (proof on next slide)

GD STEPS IN EIGENSPACE: PROOF

- A single GD step satisfies $\boldsymbol{x}^{[t+1]} = \boldsymbol{x}^{[t]} - \alpha(2\mathbf{A}\boldsymbol{x}^{[t]} - \mathbf{b})$
- Multiplying with $\mathbf{V}^T$ and subtracting $\boldsymbol{x}^*$ yields

$$\begin{aligned}
\mathbf{V}^T(\boldsymbol{x}^{[t+1]} - \boldsymbol{x}^*) &= \mathbf{V}^T(\boldsymbol{x}^{[t]} - \boldsymbol{x}^*) - \alpha\mathbf{V}^T(2\mathbf{A}\boldsymbol{x}^{[t]} - \mathbf{b}) \\
\boldsymbol{w}^{[t+1]} &= \boldsymbol{w}^{[t]} - \alpha\mathbf{V}^T(2\mathbf{A}(\boldsymbol{x}^{[t]} - \boldsymbol{x}^*) + \underbrace{2\mathbf{A}\boldsymbol{x}^* - \mathbf{b}}_{=0}) \\
&= \boldsymbol{w}^{[t]} - 2\alpha\Lambda\mathbf{V}^T(\boldsymbol{x}^{[t]} - \boldsymbol{x}^*) \\
&= \boldsymbol{w}^{[t]} - 2\alpha\Lambda\boldsymbol{w}^{[t]}
\end{aligned}$$

GD STEPS IN EIGENSPACE

Therefore, for each coordinate i:

$$
\begin{aligned}
w_i^{[t+1]} &= w_i^{[t]} - 2\alpha\lambda_i w_i^{[t]} \\
&= (1 - 2\alpha\lambda_i) w_i^{[t]} \\
&= (1 - 2\alpha\lambda_i)^{t+1} w_i^{[0]}
\end{aligned}
$$

GD ERROR IN ORIGINAL SPACE

- Move back to the original space:

$$\boldsymbol{x}^{[t]} - \boldsymbol{x}^* = \mathbf{V}\boldsymbol{w}^{[t]} = \sum_{i=1}^{d} (1 - 2\alpha\lambda_i)^t w_i^{[0]} \boldsymbol{v}_i$$

- This gives the difference in $\boldsymbol{x}$-space in closed form
- The difference vector is written in the eigenbasis, initial error components $w_i^{[0]}$ decay with rate $1 - 2\alpha\lambda_i$
- The smaller $|1 - 2\alpha\lambda_i|$ is, the faster component i decays; values close to 1 mean almost no progress per step
- Eigenvectors with larger eigenvalues typically converge fastest, especially early on

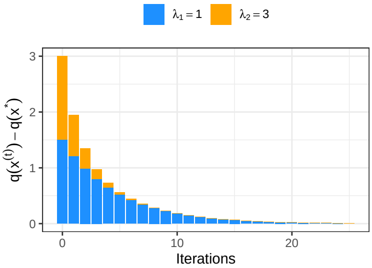
GD REGRET

- We can also analyze the regret (in objective value space):

$$q(\boldsymbol{x}^{[t]}) - q(\boldsymbol{x}^*) = \frac{1}{2}\sum_{i=1}^{d}(1 - 2\alpha\lambda_i)^{2t}\lambda_i(w_i^{[0]})^2$$

(straightforward to derive, formulas only become slightly lengthy)

- Rates become more similar once the slow modes (eigenvector components with small λ_i) dominate



OPTIMAL STEP SIZE

- Convergence requires $|1 - 2\alpha\lambda_i| < 1$ for all i
- Therefore $0 < \alpha\lambda_i < 1$
- The slowest component governs the rate:

$$\max_{i=1,\dots,n} |1 - 2\alpha\lambda_i| = \max(|1 - 2\alpha\lambda_1|, |1 - 2\alpha\lambda_n|)$$

- The optimum occurs at $\alpha = \frac{1}{\lambda_1 + \lambda_n}$
- The optimal rate equals $\frac{\lambda_n/\lambda_1 - 1}{\lambda_n/\lambda_1 + 1}$
- Convergence is driven by the condition number $\kappa = \frac{\lambda_n}{\lambda_1}$
- $\kappa = 1$ is ideal and yields convergence in one step